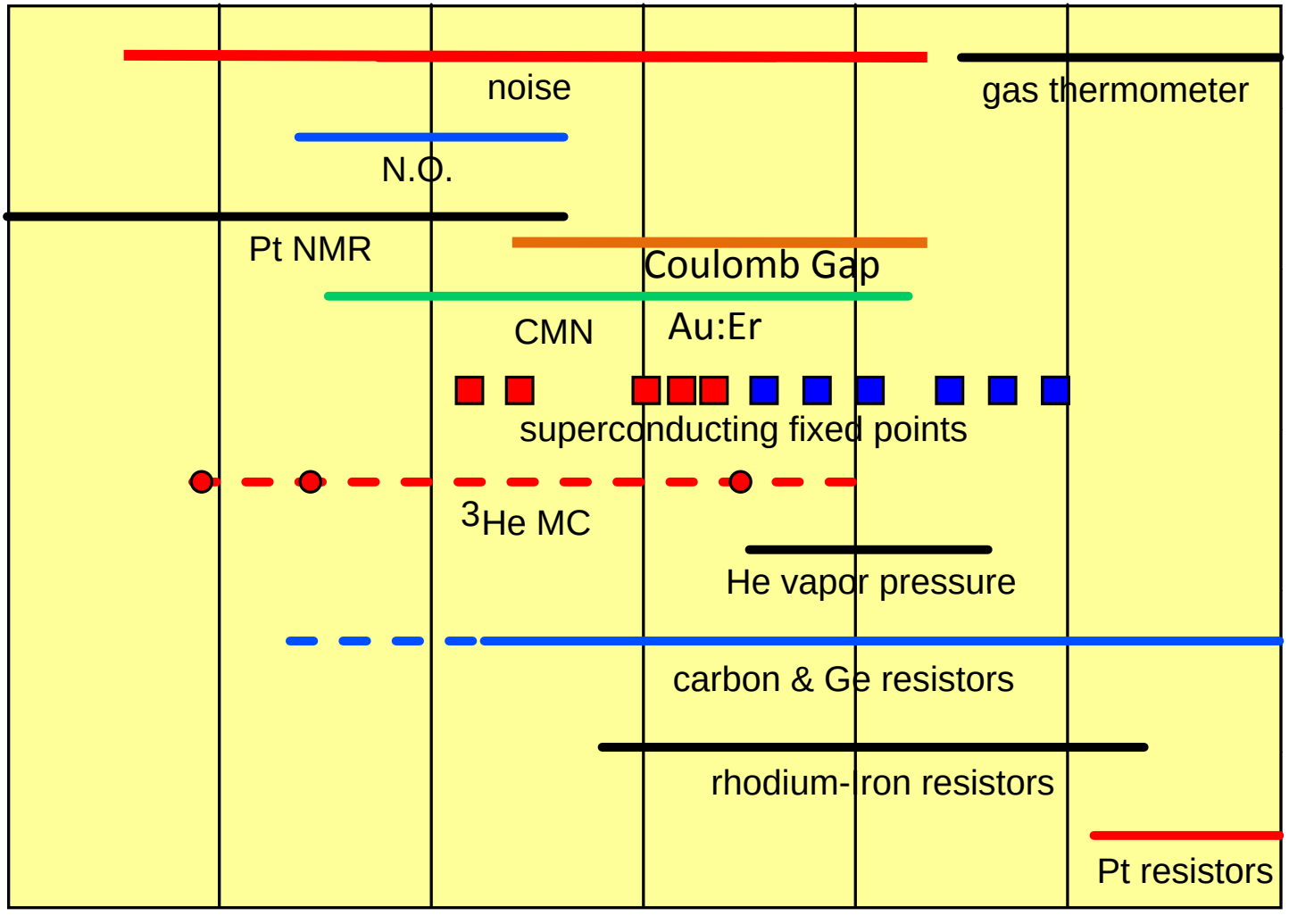
Measurent Below 1.0 K



Primary thermometersthermometers

^{3}He – meltingmelting curvecurve

Superconducting fix point devices Noise thermometry

Coulomb blockade thermometryy

Nuclear Orientation thermometry

Helium melting curve thermometer

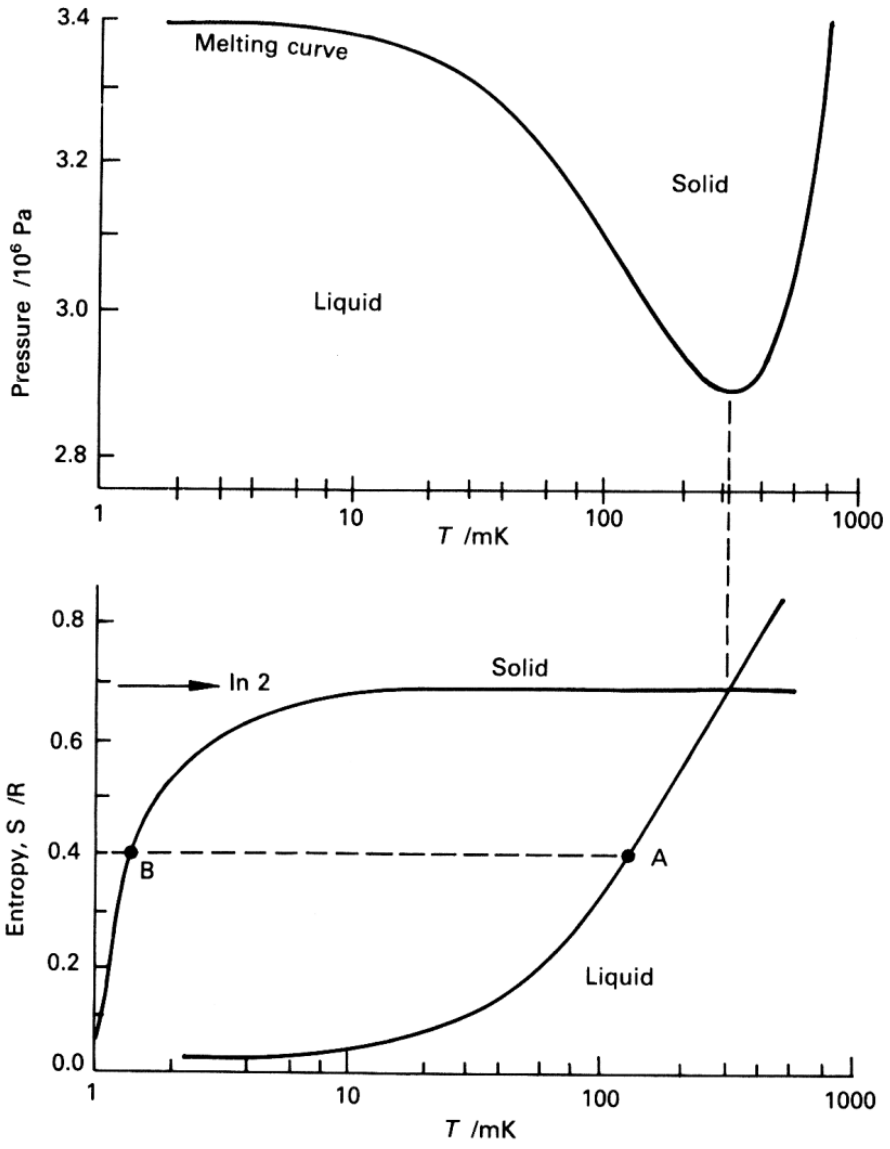


Figure 5.7 (top) *The ^{3}He melting curve* and **Figure 5.8** (bottom) *Entropy of ^{3}He in the liquid and solid phases (Betts, 1989)*

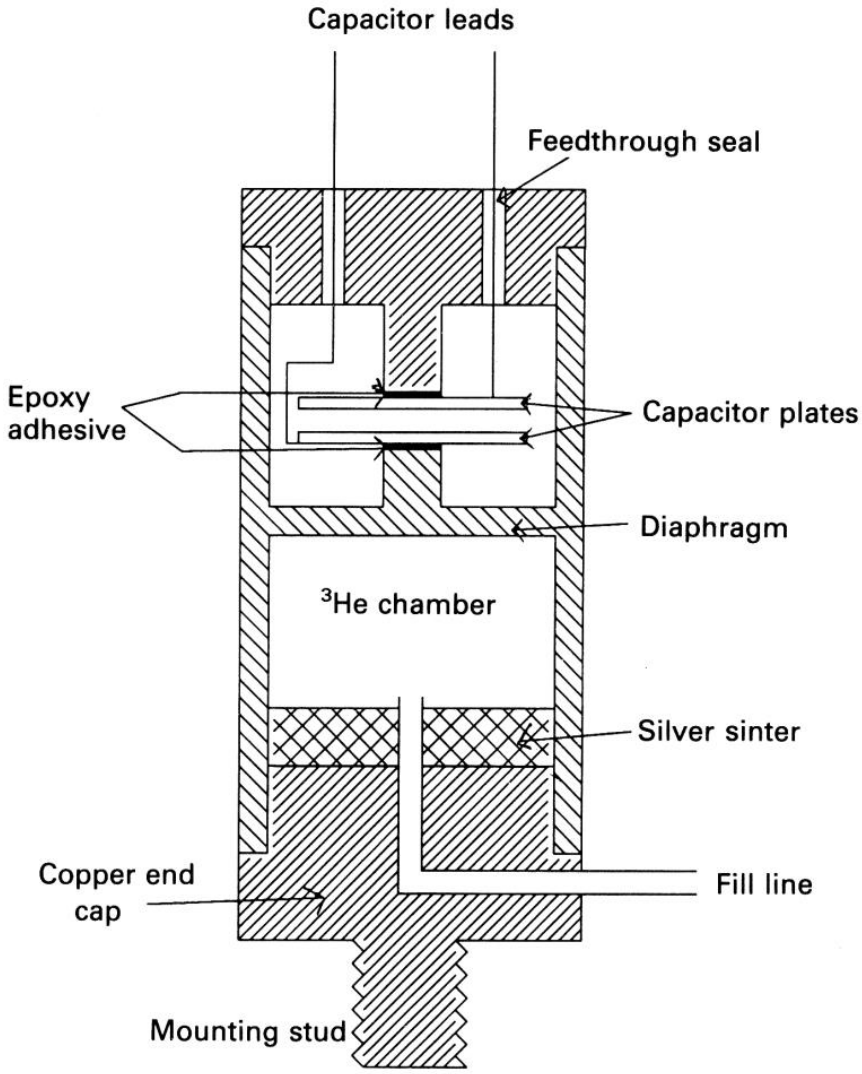
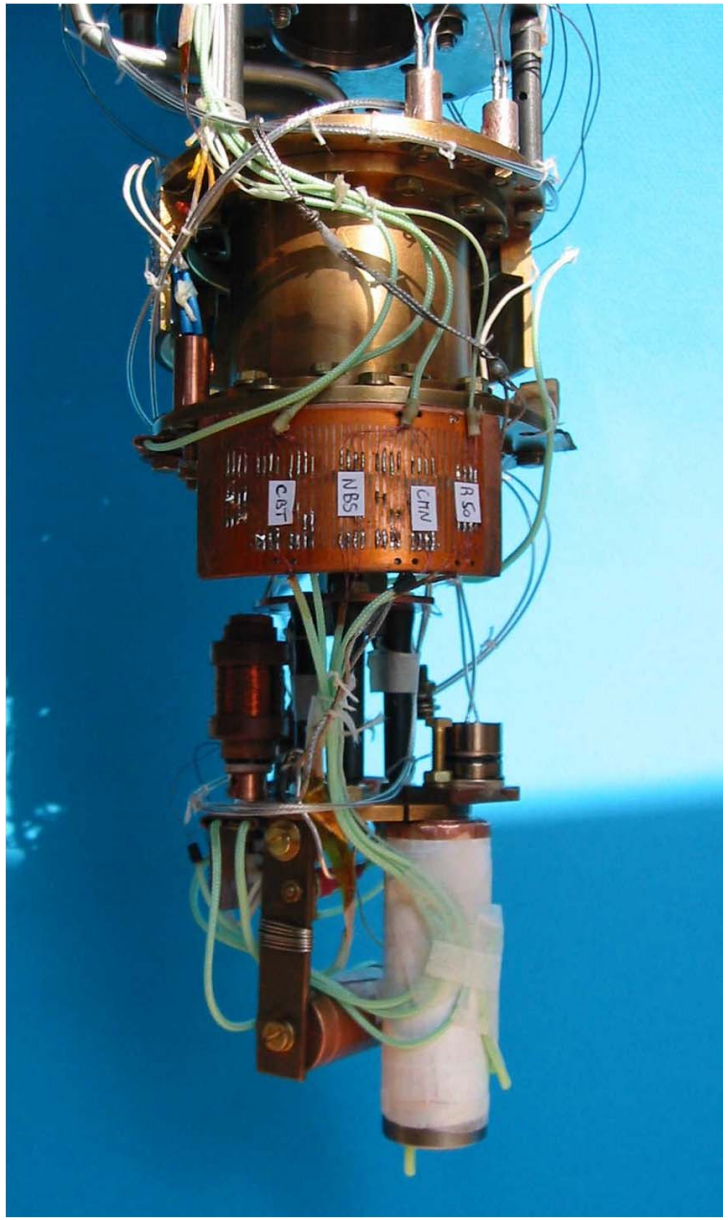
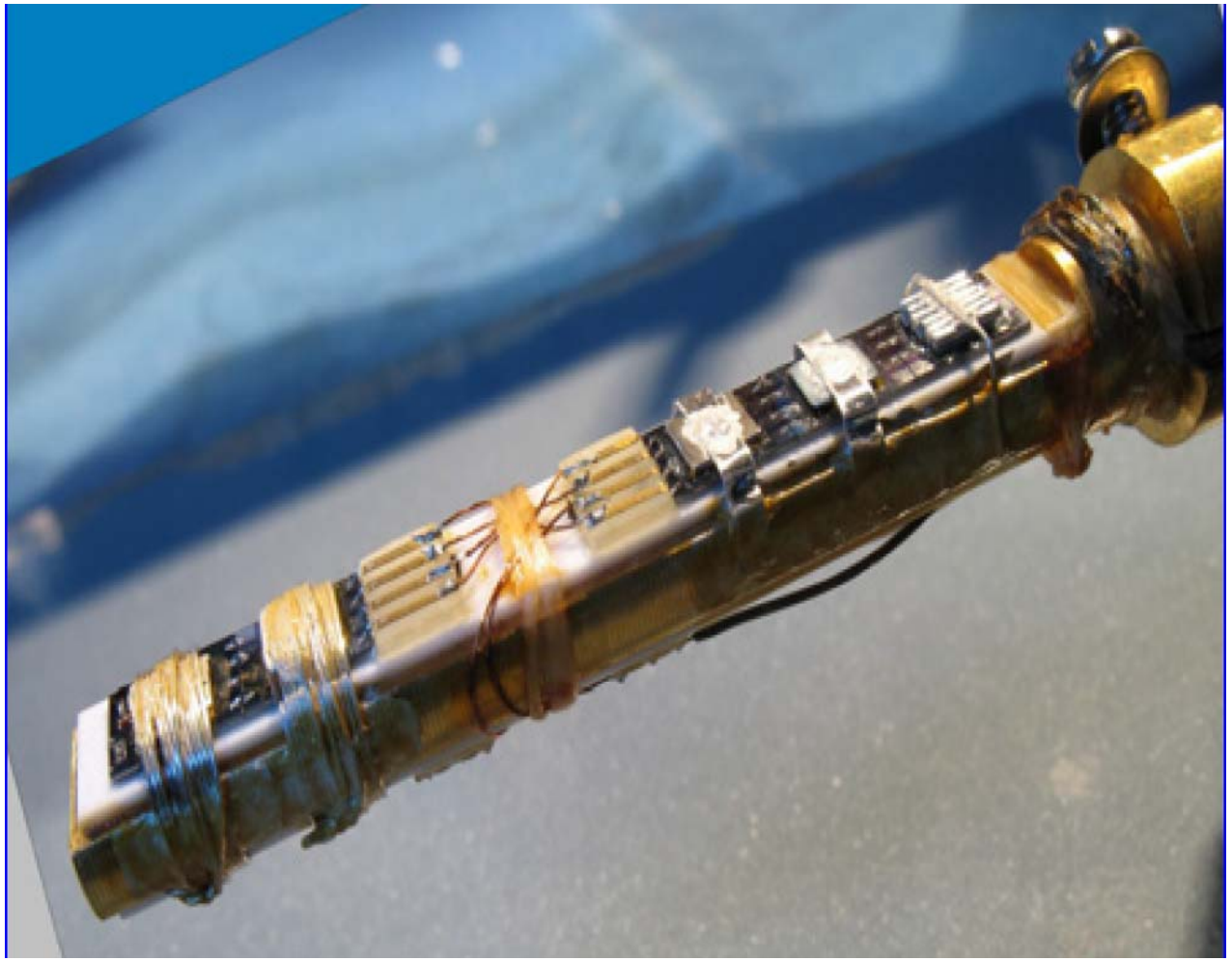


Figure 7.14 *Pressure cell suitable for ^{3}He melting curve thermometry*

Superconducting Fixpoint Devices

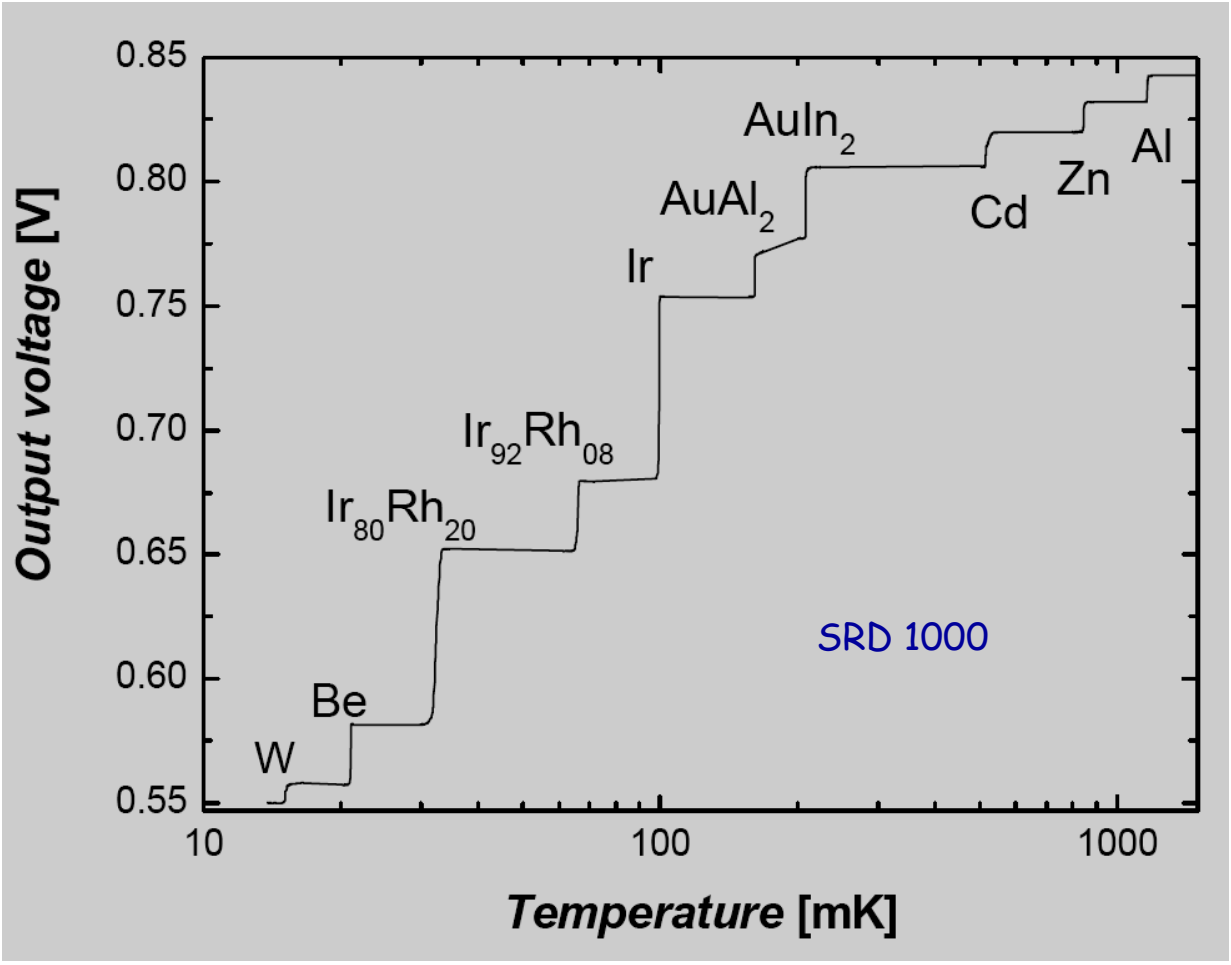
Inductive measurement of the superconducting transition





SRM 767 and SRM 768 devices made by the NIST (formerly NBS)

Superconducting Fixpoint Devices



#	material	T_C [mK]	W_C [mK]	U_C [%]
1	W	15	< 0.2	< 0.26
2	Be	21	< 0.3	< 0.28
3	$Ir_{80}Rh_{20}$	30	< 0.5	< 0.34
4	$Ir_{92}Rh_{08}$	65	< 0.5	< 0.16
5	Ir	98	< 0.5	< 0.10
6	$AuAl_2$	145	< 0.5	< 0.06
7	$AuIn_2$	208	< 1	< 0.10
8	Cd	520	< 4	< 0.16
9	Zn	850	< 3	< 0.08
10	Al	1180	< 4	< 0.06

Magnetic susceptibility

7.2.1 Magnetic susceptibility thermometry

An ideal paramagnetic material is one in which there are no interactions between the magnetic dipoles. At low magnetic field strength, the magnetic susceptibility of an ideal paramagnetic material follows Curie's law,

$$\chi = \mu_0 \frac{M}{B} = \frac{C}{T},$$

where B is the field acting on the dipoles and M the magnetisation (see Chapter 6). The Curie constant C may be calculated from first principles, so measurement of the ideal susceptibility yields absolute temperature directly. Unfortunately, at a particular value of applied field, as the temperature falls, the magnetisation increases and a point is reached where the magnetisation tends to saturate. To extend the measurement range below this point, the applied magnetic field may be reduced. The minimum value of the applied field which may be used is that which is comparable to the internal fields. This determines the minimum temperature that may be measured using paramagnetic susceptibility. The salt cerium magnesium nitrate (CMN), used for cooling by adiabatic demagnetisation, is used in paramagnetic thermometry between about 2 mK and 1 K because it has relatively low internal fields. Using dilute magnetic media in which the magnetic atoms are well separated (for example, CLMN with about 5 per cent cerium to lanthanum) reduces the problem of internal fields to some extent, but the susceptibility is smaller and, consequently, harder to measure accurately.

Figure 7.12 (top) shows schematically a simple arrangement for magnetic thermometry. The paramagnetic sample is placed within two coils, which make a transformer, and the mutual inductance between them is measured using A.C. techniques. The mutual inductance is given in terms of the

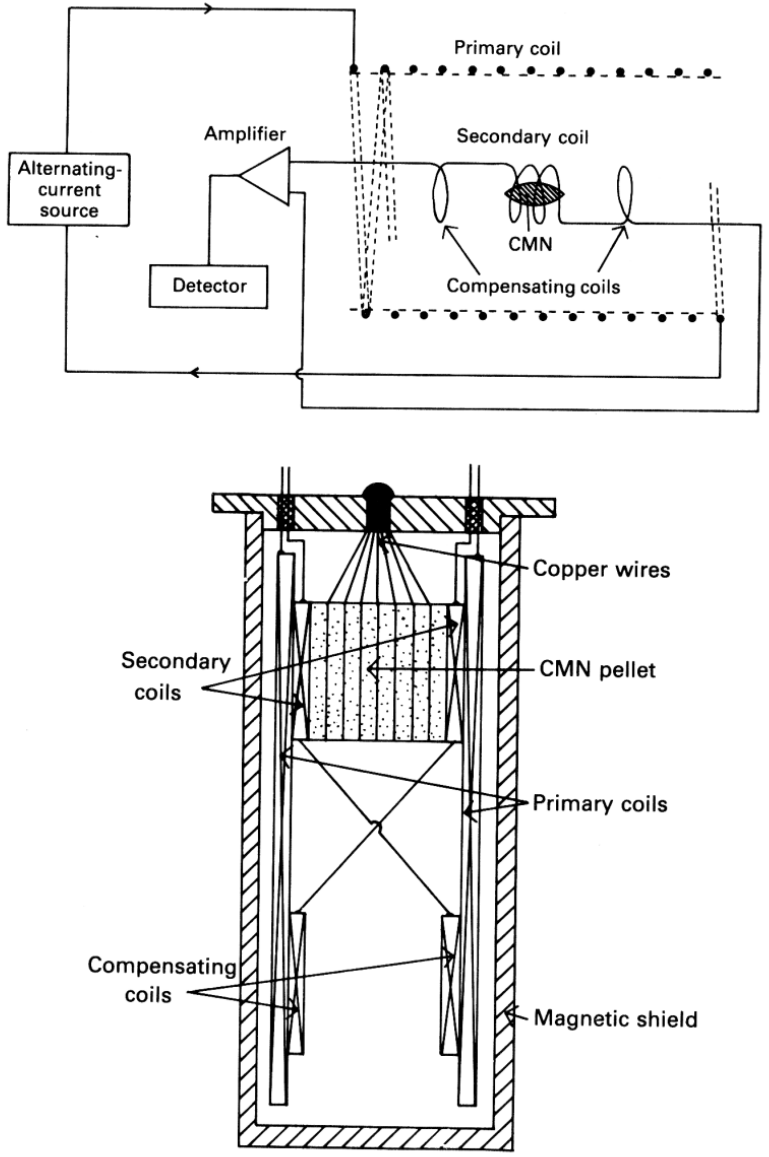
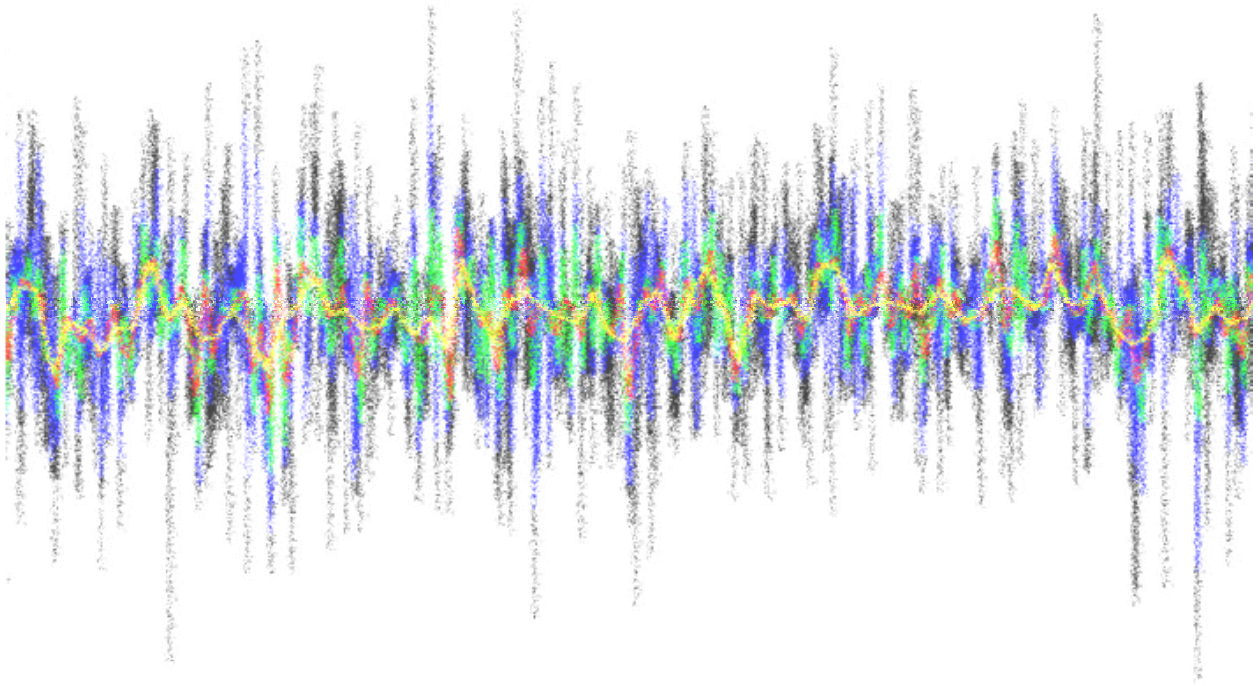


Figure 7.12 (top) *Schematic diagram of CMN magnetic susceptibility thermometer system.* (bottom) *An experimental magnetic thermometry cell*

Noise Thermometry



12.6 Noise Thermometry

The conduction electrons in a metal perform random thermal movements (*Brownian motion*), which result in statistical voltage fluctuations of a resistive element. Therefore all resistive elements of an electronic circuit are noise sources. This noise is statistical, therefore we cannot make statements about its value at a fixed time, and the mean value of the noise voltage vanishes. However, we can calculate an effective time averaged mean square noise voltage

$$U_{rms} = \sqrt{\langle u^2 \rangle_t} \, . \tag{12.10}$$

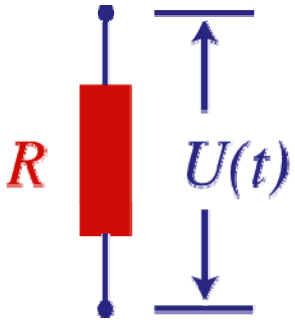
This noise voltage was investigated in 1928 by J.B. Johnson (*Johnson noise*) and H. Nyquist (*Nyquist theorem*). Nyquist arrived at the following equation for the component of the time averaged noise voltage within the frequency band from ν to $\nu + d\nu$ of a resistor with value R at temperature T:

$$\langle u^2(\nu) \rangle_t = 4k_B T R d\nu \, . \tag{12.11}$$

If we measure within a frequency band of width $\Delta\nu$ we obtain

$$\langle u^2 \rangle = \int_{\nu}^{\nu+\Delta\nu} 4k_B T R d\nu = 4k_B T R \Delta\nu \, . \tag{12.12}$$

Johnson & Nyquist (1928)



Thermal Fluctuations of the voltage across a conductor

$$S_U = \frac{\langle U^2 \rangle}{\Delta f} = 4 k_B T R$$

Quantum corrections:

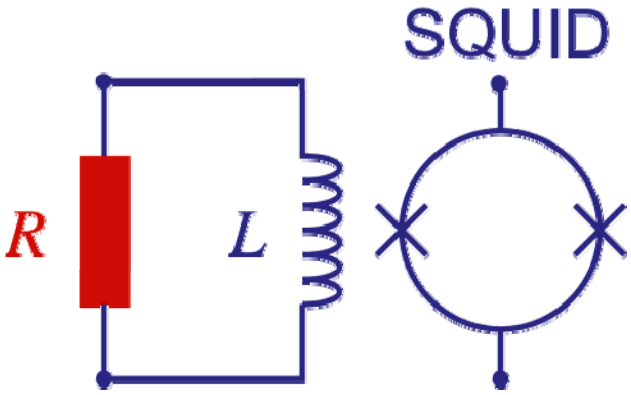
$$S_U = 4hfR \left[\frac{1}{2} + \frac{1}{e^{hf/k_B T} - 1} \right]$$

$$\simeq 4 k_B T R \left[1 + \frac{1}{12} \left(\frac{hf}{k_B T} \right)^2 \right]$$

can be neclegted since
(T > 100 mK, f < 1 kHz)

$$\frac{hf}{k_B T} < 5 \times 10^{-4}$$

Current Noise



Current noise if shorted

$$S_I = \frac{4k_\mathrm{B}T}{R}$$

Sensitivity of a current sensing

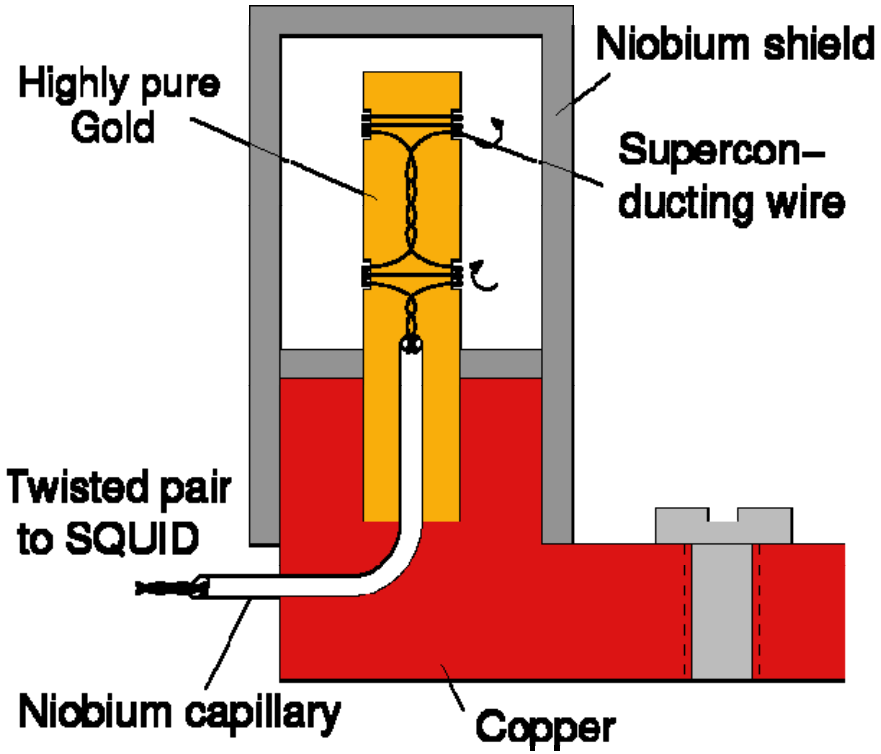
DC SQUID < 1 pA/√Hz

$$S_I = \frac{4k_\mathrm{B}T}{R}\frac{1}{1+(f/f_0)^2}$$

$$f_0 = \frac{1}{2\pi}\frac{R}{L}$$

$$\overline{E} = \int_0^\infty \frac{1}{2} L\, S_I \,\mathrm{d}f = \frac{1}{2}k_\mathrm{B}T$$

Inductively Read Out Noise Thermometer

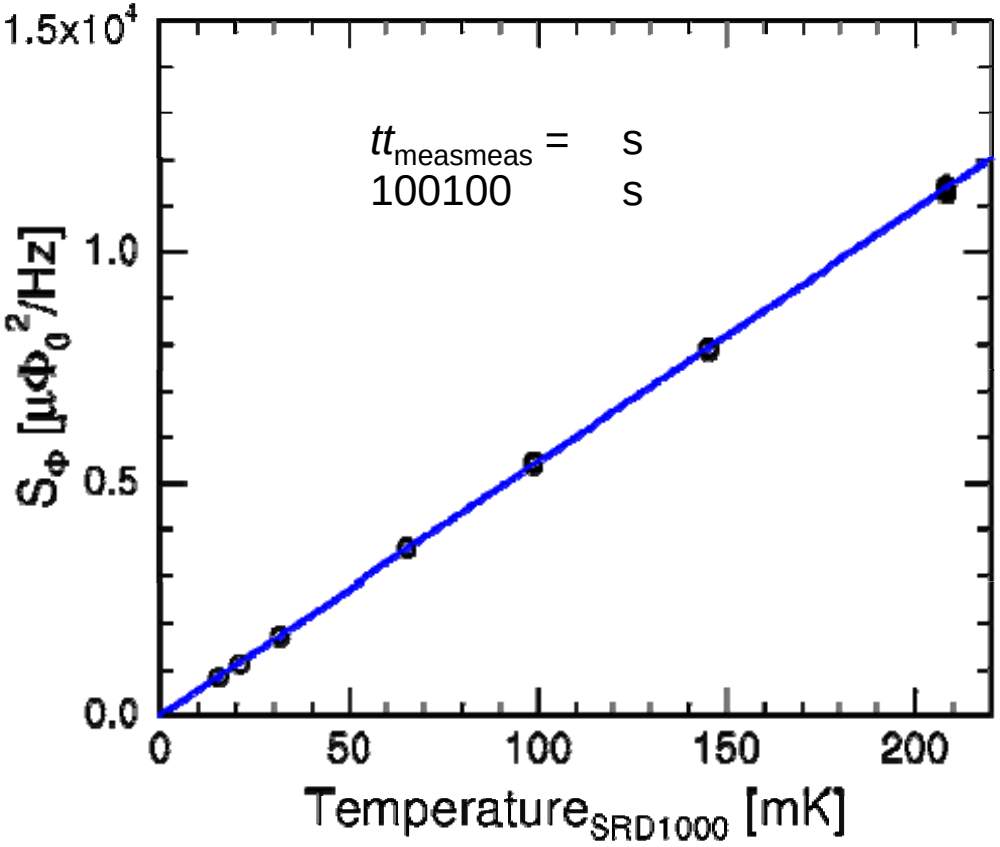


Noise source :

Gold cylinder ∅ 2 mm, purity > 99,999%, RRR = 110

Kupfer cylinder, ∅ 2,5 mm, purity > 99,999%, RRR = 1000

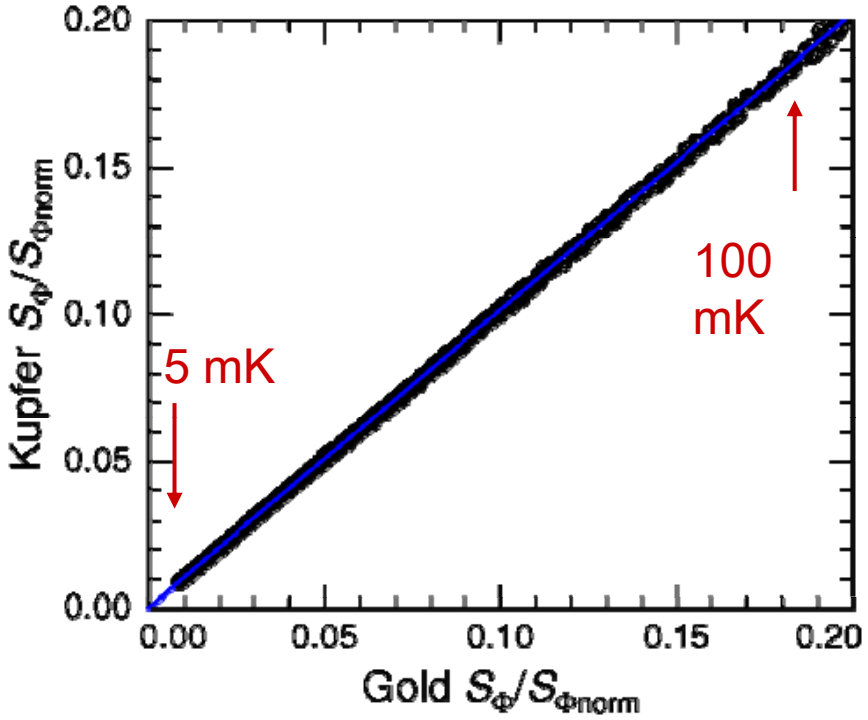
Noise thermometer – Fixpoint device



Linear temperature dependence of the noise spectral power

$$S_\Phi \sim T$$

Copper & Gold



Deviations from expected linear
Dependence small than 0,5 %

Copper-Thermometer: RRR = 1000
Gold-Thermometer: RRR = 110

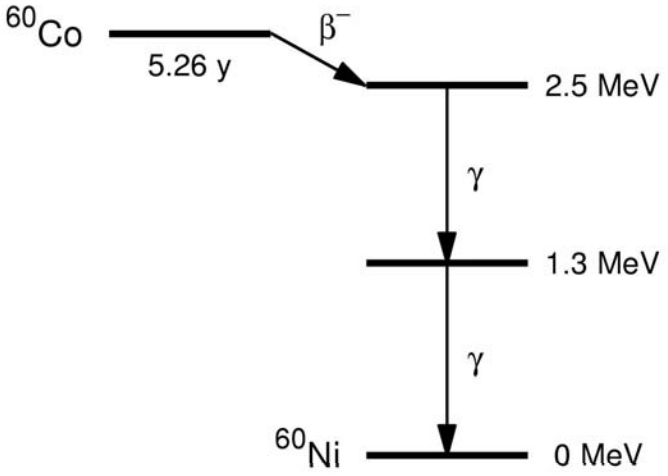
Nuclear orientation thermometry (1-40 mK) y-ray anisotropy Co- 60 Thermomenter

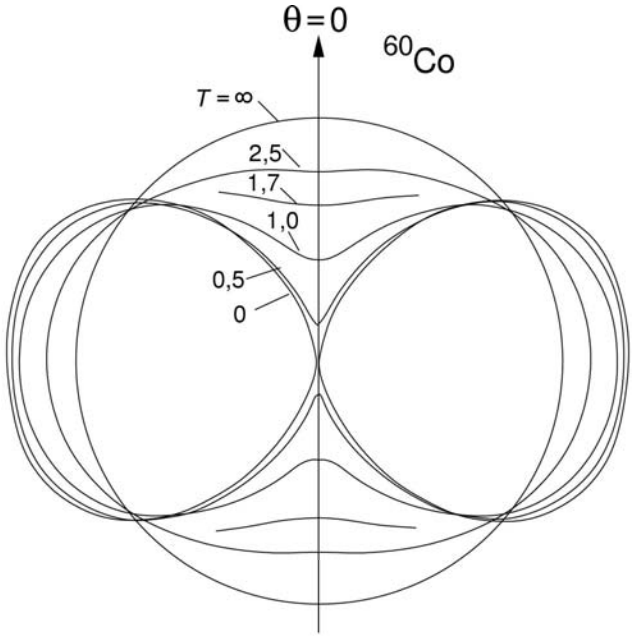
- Important as primary thermometer
  - We use it to calibrate more sensitive thermometers
- Thermometer is based on measuring the anisotropy of γ-rays emitted from polarized nuclei
  - Nuclear levels are split by magnetic field into $(2I+1)$ hyperfine levels (I is the spin quantum number of nucleus)
  - The population numbers of the sublevels follow Boltzmann statistics and so depend on temperature
  - On the other hand orientation of nuclei vary from level to level -> temperature depended angular radiation pattern $W(\phi, T)$
- Sensitivity is reasonable between 3 and 30 mK, at higher temperatures all energy levels are equally populated and at lower temperatures only the lowest level is populated.
- Measuring one spectrum takes roughly one hour

Nuclear Orientation Thermometer

example ^{60}Co

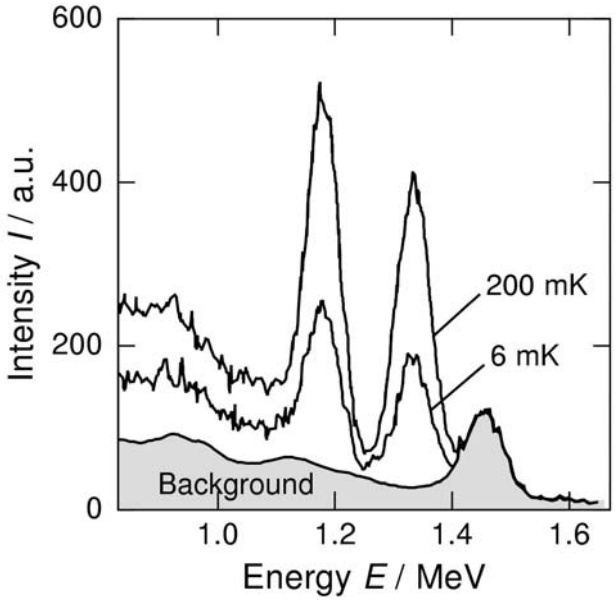
nuclear spin I
magnetic field: splitting into (2I+1) sublevels

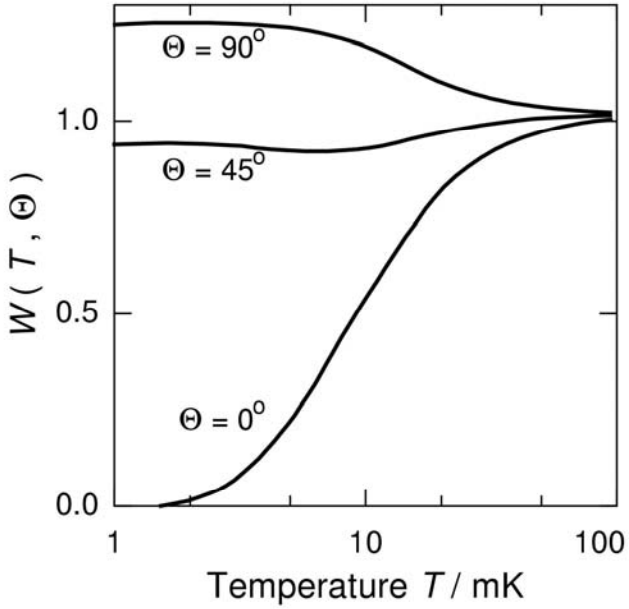




Nuclear Orientation Thermometer

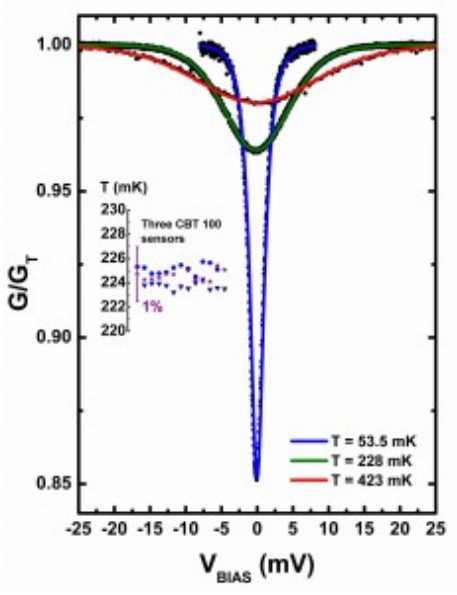
$$W(T,\Theta) = 1 + \sum_{k=2,4,\ldots}^{k_{\max}} G_k\, U_k\, F_k\, B_k(T)\, P_k(\cos\Theta)$$





Coulomb blockade thermometer

CBT is based on change of electric conductance of tunnel junction arrays. In CBT, the differential conductance of tunnel junction array is a bell-shaped curve.



Normalized conductance at three temperatures.

The advantages of CBT are
- primary thermometer does not need external calibration - secondary mode for fast measurement of zero-bias resistance and temperature
- insensitivity to magnetic field
- simple conductance measurement using lock-in amplifier

Disadvantages:
- Limited temperature range 10 mK – 200 mK - slow

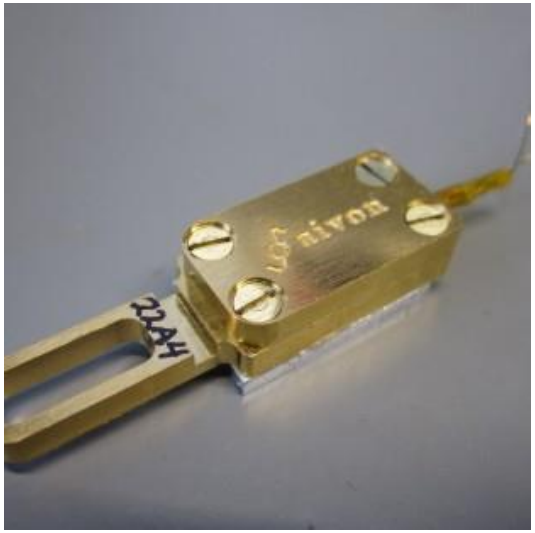
It has been shown [1], that the full-width at half maximum ($V_{1/2}$) of the curve depends only on the (electronic) temperature of the sensor and on some constants of nature.

$V_{1/2} = 5.439\ N\ k_B T/e$, (1)

where N is the number of tunnel junctions in the array, Boltzmann constant k_B = 1,38 exp(-23), T is the temperature and electron charge e=1,609 exp(-19).

Equation (1) forms the basis for primary thermometer since temperature can be calculated when the width of the conductance curve is measured. For faster measurements one can monitor only the depth of conductance dip at zero bias which depends on capacitance and temperature. The initially unknown capacitance can be calibrated once the temperature is measured using equation (1).

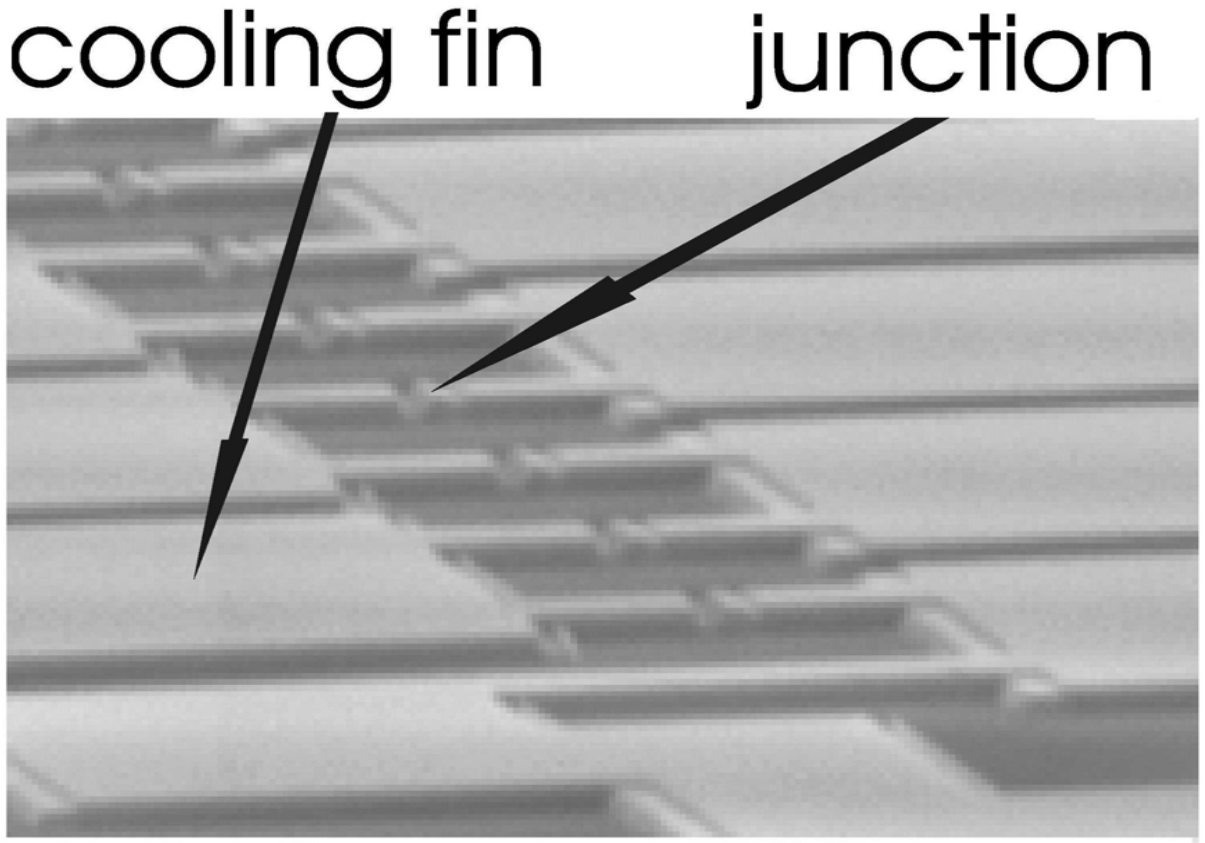
[1] J. Pekola et al, Physical Review Letters, 73, 2903 (1994)]



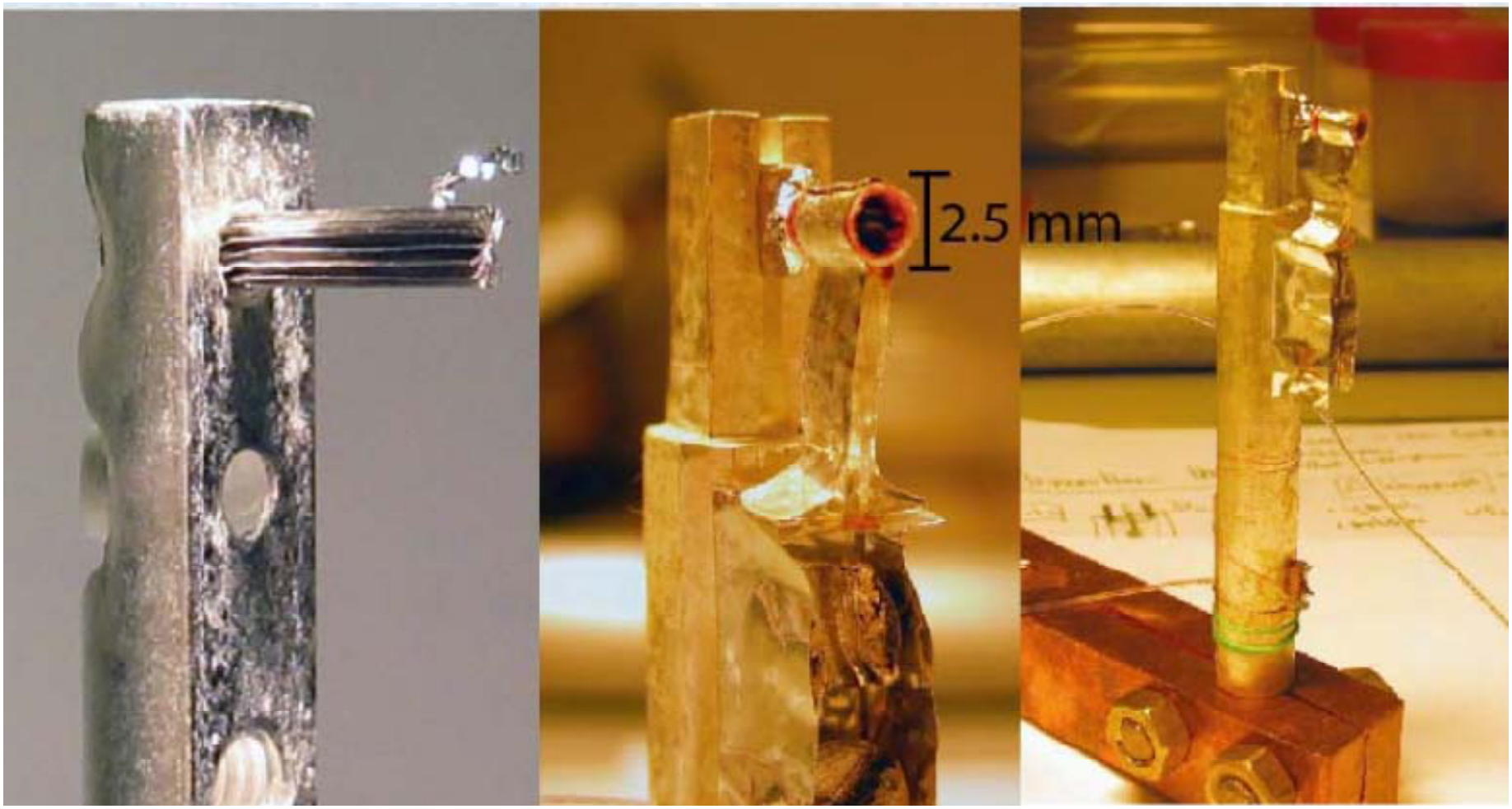
http://www.aivon.fi/

Coulomb Blockade Thermometer

Conductance through an array of tunnel junctions



Pt-NMR-Thermometry



(Pt) NMR thermometer

Sample is placed in a static magnetic field B_0

Susceptibility of nuclear spins follows the Curie law

$$\chi_n = \Lambda_n / T$$

We measure the magnetization M_n by applying an excitation pulse at the Larmor frequency to tip the magnetization by a small angle θ

After excitation pulse there is a component

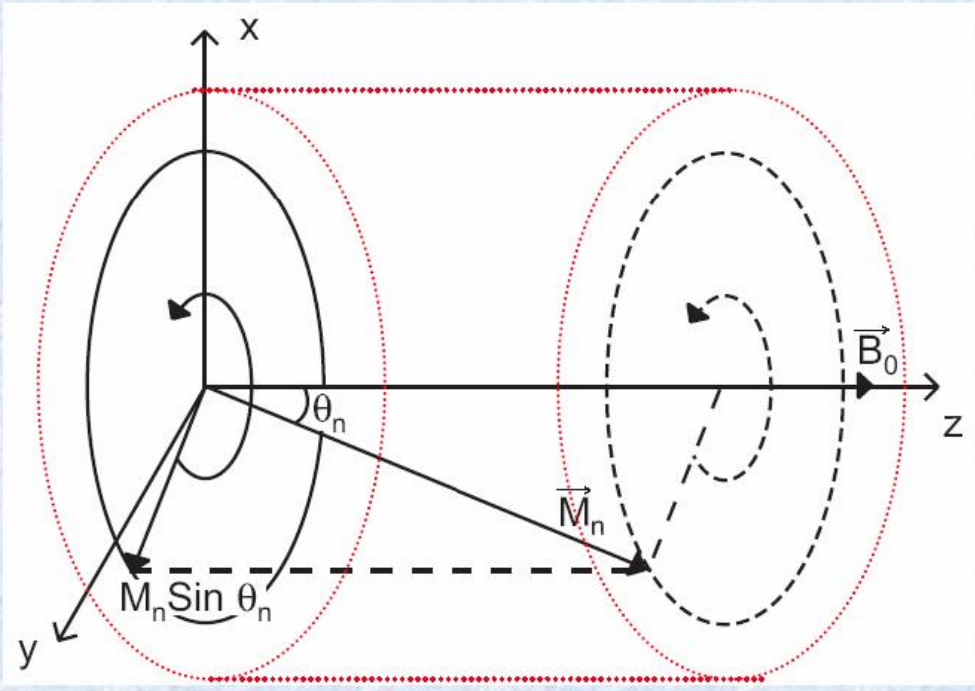
$$B_y = B_1 sin(\omega_n t)$$

precessing in the XY-plane

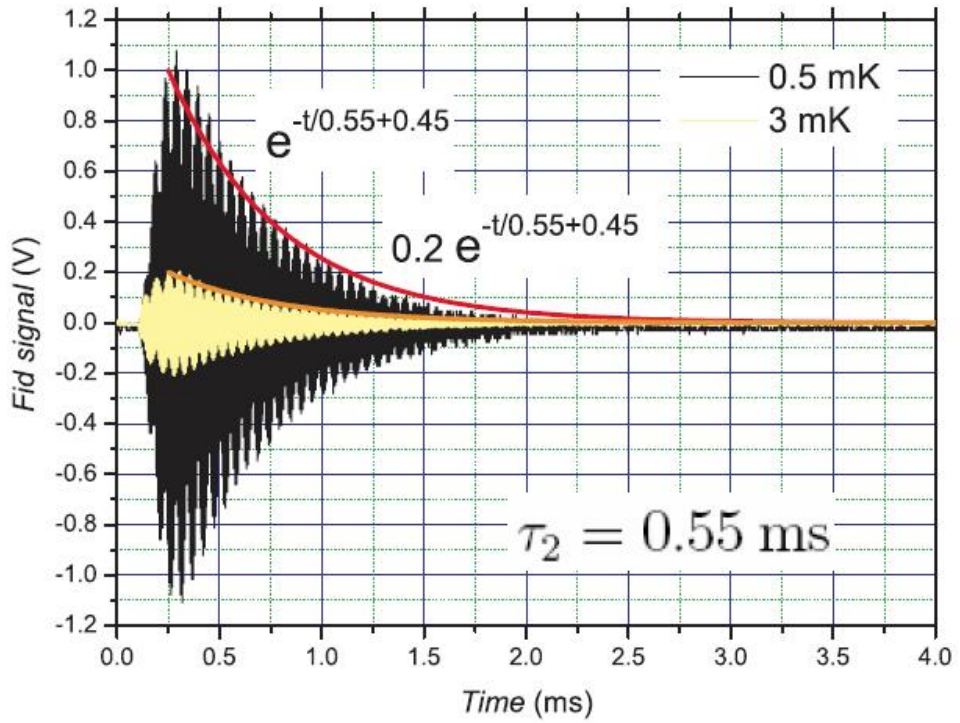
This component induces a voltage

$$U = \alpha \omega_n M_n sin\theta_n,$$

to pick-up coil.

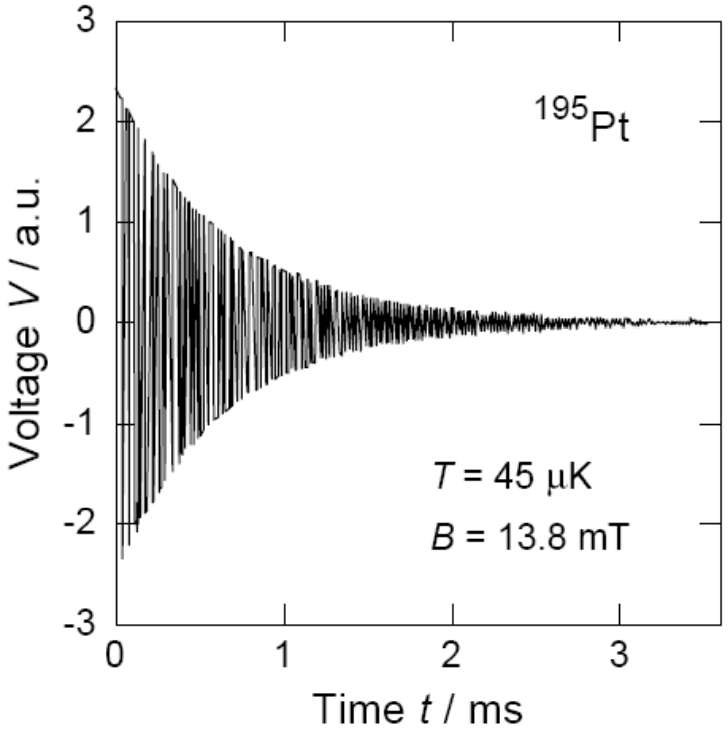


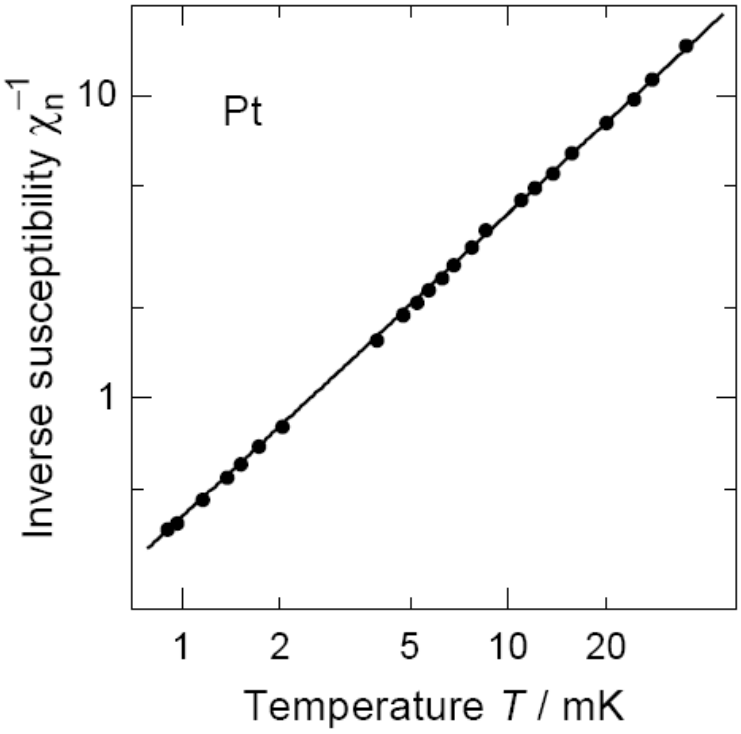
•This signal is called the free induction decay and it's amplitude is directly proportional to magnetization and thus inversely proportional to temperature



- Two important timescales
  - τ_1 determines nuclear spin systems relaxation time to electron system's temperature $\tau_1=\kappa/T_e$, where κ is Korringa constant (0.030 sK for Pt, τ_1 ~ 5 min at 100 μK)
  - τ_2 determines spin-spin relaxation time. Sufficient time is needed for measurement to be made. For Pt τ_2 ~ 1 ms
- In addition Pt has only one magnetic isotope ^{195}Pt with nuclear spin ½ -> simplifies signal
- Pt is a good sample material

Pt-NMR-Thermometry





$$M_y(t) = \frac{1}{2} M_0 \cos(\Omega t)\, \mathrm{e}^{-t/\tau_2^*}$$

mixing chamber of the dilution refrigerator

copper plate and melting curve thermometer

experimental cell (NMR of liquid 3He)

3He NMR thermometer

